

SUMMARY

I am a 3rd-year PhD candidate in MAMBO team at Toulouse University (expected graduation: Dec. 2026), advised by [Prof. Pierre Weiss](#) and [Prof. Edouard Pauwels](#). My research focuses on computational imaging with deep learning and generative models. I also actively contribute to open-source software.

WORK EXPERIENCE

Research Engineer – [Agenium Space](#) Dec 2022 - Sept 2023

Correcting micro-vibrations in satellite imaging with pushbroom cameras. Modeling the imaging process, implementing deep learning-based methods, and evaluating them on real data from Sentinel-2B.

Research Intern – [Toulouse Mathematics Institute](#) Jun 2022 - Sept 2022

Product-convolution neural networks: applications to image deblurring with space-varying blurs, with [Pierre Weiss](#) and [Paul Escande](#).

EDUCATION

2023 - present PhD (Applied Mathematics, Inverse problems in Imaging) at **Toulouse University**
2018 - 2023 Master of Engineering in Applied Mathematics at **INSA Toulouse** (Major in AI / Data Science)

OPEN-SOURCE PROJECTS

Blind deblurring in Fluorescence Microscopy Napari plugin released soon!

Designing deep learning-based methods to jointly estimate the sharp image and the unknown blur kernel from a single blurred fluorescence microscopy image. We address diffraction-limited blurs based on Zernike polynomials, spatially varying blurs and 2D/3D settings. Efficient implementation in PyTorch and ongoing validation on various microscopy settings (RIM, TIRF, confocal). Joint work with [Pierre Weiss](#), [Florian Sarron](#), [Paul Escande](#) and opticians [Thomas Mangeat](#), Sylvain Cantaloube. Presented at [MIFOBIO-25](#).

DeepInverse – Core contributor and maintainer [Link to Github](#)

An open-source PyTorch-based library (part of the [PyTorch Ecosystem](#)) for solving imaging inverse problems with state-of-the-art deep learning based methods. `deepinv` accelerates research across imaging domains, enhances research reproducibility via a common modular framework of problems and algorithms. Responsibilities: core library API design/contribution, benchmarks, CI/CD, packaging, docs, and community support.

SKILLS

Programming	Python: PyTorch (advanced), triton/cuda kernel (intermediate), JAX. C++ (intermediate) Profiling/optimization (Nsight, torch-profiler)
Development	Git/Github, packaging (PyPI), virtual environments (<code>conda</code> , <code>uv</code> , <code>docker</code>). SOTA implementation and evaluation, model training and fine-tuning.
Research	Image restoration, computational imaging, deep learning, deblurring/denoising
Language	English (professional), French (fluent), Vietnamese (native)

TEACHING

• Teaching Assistant (Vacataire) – **INSA Toulouse** 2023 - present
Courses: Signal Processing, Stochastic Optimization (32 hours per year)

DISTINCTIONS

- French National Prize for Open Science Library for DeepInverse Nov 2024
- Doctoral Research Fellowship, French Ministry of Higher Education and Research 2023 - 2026
- IA Pau Winner 2023

PUBLICATIONS

Nguyen, Minh Hai, Quoc Bao Do, et al. (2026). *An analytic theory of convolutional neural network inverse problems solvers*. arXiv: [2601.10334](https://arxiv.org/abs/2601.10334) [cs.CV]. URL: <https://arxiv.org/abs/2601.10334>.

Nguyen, Minh-Hai et al. (2025). *How Diffusion Prior Landscapes Shape the Posterior in Blind Deconvolution*. arXiv: [2508.02923](https://arxiv.org/abs/2508.02923) [cs.CV]. URL: <https://arxiv.org/abs/2508.02923>.

Tachella, Julián et al. (2025). “DeepInverse: A Python package for solving imaging inverse problems with deep learning”. In: *Journal of Open Source Software* 10.115, p. 8923. DOI: [10.21105/joss.08923](https://doi.org/10.21105/joss.08923). URL: <https://doi.org/10.21105/joss.08923>.

Nguyen, Minh Hai and Pierre Weiss (2024). “Comparing Plug-and-Play and Unrolled networks”. In: *Pre-print*.

Nguyen, Minh Hai et al. (2024). “DeepVibes: Correcting Microvibrations in Satellite Imaging With Pushbroom Cameras”. In: *IEEE Transactions on Geoscience and Remote Sensing* 62, pp. 1–9.